

Muhammed Irfan K M

Software Developer | Full-Stack & AI/ML

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PROFESSIONAL SUMMARY

Artificial Intelligence & Machine Learning undergraduate with project experience in full-stack development, backend systems, computer vision, and deep learning. Strong foundation in Data Structures & Algorithms, Python, C, JavaScript, SQL, React, Node.js, REST APIs, PyTorch, and OpenCV. Experienced in developing practical applications, integrating APIs, designing databases, and applying machine learning to real-world problems.

EDUCATION

Sahyadri College of Engineering & Management

2023 – 2027

Bachelor of Engineering in Artificial Intelligence & Machine Learning

Mangaluru, India

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, Computer Vision, Artificial Intelligence

TECHNICAL SKILLS

Programming Languages: Python, C, JavaScript, TypeScript, SQL

Software Development: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Full Stack Development, Problem Solving

AI/ML: Machine Learning, Deep Learning, Computer Vision, CNN, YOLOv8, TensorFlow, PyTorch, OpenCV

Frameworks & Libraries: React, Next.js, Node.js, Express.js, NumPy, Pandas, Scikit-learn, Gradio

Databases & Authentication: MySQL, MongoDB, Firebase, JWT, bcrypt

Developer Tools: Git, GitHub, VS Code, Google Colab, Postman

Languages: English (Fluent), Malayalam (Fluent), Hindi (Fluent), Arabic (Basic)

PROJECTS

IRFAN OS – Interactive Developer Portfolio | *Next.js, TypeScript, React Three Fiber, Three.js, GSAP*

2026

- Built an immersive operating system-inspired developer portfolio featuring interactive 3D environments, terminal navigation, AI assistant, voice commands, mini-games, and achievement systems.
- Integrated interactive UI modules, live GitHub data, responsive layouts, and multiple application modes into a single web experience.

Portable Microplastic Screening Using Deep Learning | *Python, YOLOv8, PyTorch, OpenCV, Roboflow, Gradio*

2025 – 2026

- Developed a computer vision system for detecting microplastics from optical microscopy images using YOLOv8 and PyTorch.
- Integrated OpenCV-based processing, a Roboflow dataset, Gradio inference interface, and a safety scoring layer.
- Designed the system to provide interpretable detection and risk information from microscopy-based analysis.

MedQR Care – Emergency Medical QR Platform | *React, Node.js, Express.js, MongoDB, JWT*

2026

- Built a full-stack emergency medical ID platform providing QR-based access to essential patient information through a public lookup system.
- Implemented JWT authentication, REST APIs, MongoDB data management, rule-based risk scoring, and rate limiting for public access.
- Separated authenticated patient operations from minimal public emergency data access using unique, rotatable identifiers.

Expense Splitter | *Node.js, Express.js, React, MySQL, JWT*

2025

- Built a full-stack expense management application with JWT authentication and RESTful APIs.
- Implemented group expense tracking and balance settlement logic backed by MySQL.
- Structured the application with separate frontend and backend components for maintainable full-stack development.

LEADERSHIP & ACTIVITIES

Hackathon Participant

2024 – Present

Academic Projects & Hackathons

- Participated in 3 hackathons involving software, AI, and application-oriented problem solving.
- Collaborated with team members to develop and present technology-based solutions within limited development timelines.
- Applied programming, problem-solving, and communication skills while working under competitive development constraints.

CERTIFICATIONS

Infosys Springboard – Python and Programming Fundamentals, Artificial Intelligence